

Finite Realization of Coarse-Information Profiles

Countable Turing ideals, generic secret sharing,
and simultaneously unattained representative spectra

Prepared for Vladimir Reshetnikov with ChatGPT

28 September 2026

Abstract

For a binary sequence X , its coarse-information core consists of the sets computed by every density-zero perturbation of X . We prove a finite realization theorem: an assignment of countable Turing ideals to coalitions of finitely many participants is realizable as the cores of their oracle joins exactly when it is monotone and assigns the computable ideal to the empty coalition. All nonempty coalition classes can simultaneously be made to have no least Turing-degree representative. Relative to a presentation P of the ideals, the full join H can satisfy $P <_T H <_T P'$ and $H' \equiv_T P'$.

The argument first verifies the single-ideal construction already sketched in the repository's target C4, then combines robust column coding with finite additive sharing and a generic-perturbation lemma. A coalition normal form proves the essential upper bound: its core contains no unintended information. As a particularly sharp consequence, two sequences with computable cores can have any prescribed countable Turing ideal as the core of their join. A separate access-structure version makes every unauthorized coalition itself 1-generic relative to the secret presentation.

The non-elementary inputs are named published theorems of Hirschfeldt, Jockusch, Kuiper, and Schupp. The finite-profile results are presented with conventional proofs as proposed research contributions, not as externally certified priority claims or machine-verified theorems. A proof audit, finite verification program, and further research agenda accompany the text.

Scope. This article does not claim a new solution to the already-retired least-representative question C1, nor a solution to the repository's question about *minimal* elements of coarse spectra. Its main object is simultaneous control of all finite coalition cores.

[write] **Report 11 of the CoarseDegrees programme.** This is the delivered manuscript, filed as research report 11 of Computability/TuringDegrees/Research/CoarseDegrees/research-reports/ (batch 37, manuscript 07). Text marked [write] was added then; the provenance, a notation dictionary against the research plan, the formal status and the plan targets addressed are in Sections 2.4–2.7. Nothing in this report is formalized. It is AI-assisted and unrefereed.

Contents

1	The problem and the main theorem	3
2	Repository provenance and dependency boundary	4
3	Basic density and oracle calculus	8
4	Generic coordinates and a transfer principle	10
5	Verifying C4: one countable ideal as a core	12
6	Two masks reveal an arbitrary ideal	14
7	Finite additive sharing and coalition normal forms	15
8	Proof of simultaneous realization	18
9	Representative spectra, nonattainment, and jump bounds	19
10	Access structures with genuinely generic unauthorized views	21
11	A boundary: leastness cannot be prescribed arbitrarily	23
12	Proof audit and reproducible finite verification	23
13	Further research questions and topics	26
14	Conclusion	28
A	Notation and construction index	29
B	Source review and reproducibility details	29

Reading guide

For the classification statement, read Section 1. For the shortest substantial example, read the two-share construction in Section 6. Its complete upper-bound mechanism is the transfer lemma in Section 4.

The proof of the general theorem is concentrated in Sections 7–9: finite sharing, the coalition normal form, elimination of unintended information, nonattainment, and the jump calculation. Section 10 gives the stronger genericity property for unauthorized coalitions. Section 11 explains why positive leastness flags cannot be prescribed without further conditions.

The source and dependency distinctions are in Section 2; the audit and finite-check results are in Section 12. The questions in Section 13 are separate research targets, not additional assertions established by the main theorem.

1 The problem and the main theorem

1.1 Information surviving arbitrary sparse errors

Think of a set of natural numbers as an infinite binary oracle. Its exact Turing degree measures everything obtainable from a completely accurate copy. A different question is what remains obtainable when an adversary may change the oracle on an unknown set of asymptotic density zero. The reduction used to recover a particular set is allowed to depend on the corrupted copy. This nonuniformity is essential.

For $E \subseteq \mathbb{N}$ and $N \geq 1$, put

$$\rho_N(E) = \frac{|E \cap [0, N]|}{N}, \quad \delta(X, Y) = \limsup_{N \rightarrow \infty} \rho_N(X \triangle Y).$$

Here $X \triangle Y$ is symmetric difference, equivalently bitwise exclusive or. It is not the oracle join $X \oplus Y$. Write $D \approx X$ when $\delta(D, X) = 0$, and let

$$\text{CD}(X) = \{D \in 2^{\mathbb{N}} : D \approx X\}, \quad \text{Core}(X) = \{A : (\forall D \in \text{CD}(X)) A \leq_T D\}.$$

This is the invariant denoted X^c in Hirschfeldt–Jockusch–Kuyper–Schupp [1]. A *Turing ideal* is a nonempty collection of sets closed downward under Turing reducibility and under finite oracle joins. Let Comp be the ideal of computable sets and $\text{Low}(B) = \{A : A \leq_T B\}$ the principal ideal generated by B .

For participants $[n] = \{1, \dots, n\}$, set

$$X_C = \bigoplus_{i \in C} X_i \quad (C \neq \emptyset), \quad X_\emptyset = \emptyset.$$

All finite joins in this article use equal-density periodic interleaving, in a fixed ordering of their coordinates. In particular, X_C is a specified oracle, not merely an unspecified representative of a join degree.

Theorem 1.1 (Finite profile realization). *Let $n \geq 1$. An assignment $(I_C)_{C \subseteq [n]}$ is of the form*

$$I_C = \text{Core}(X_C) \quad (C \subseteq [n]). \tag{1.1}$$

for some binary sequences $X_1, \dots, X_n$ if and only if:

- (i) *every I_C is a countable Turing ideal;*
- (ii) $I_\emptyset = \text{Comp}$;
- (iii) $I_C \subseteq I_D$ *whenever $C \subseteq D$.*

Moreover, the realizing sequences can be chosen so that every nonempty C has no least Turing degree in its density-zero class, its uniform coarse-equivalence class, or its nonuniform coarse-equivalence class.

Choose sequences $(A_{S,k})_{k \in \mathbb{N}}$ generating I_S for all nonempty $S \subseteq [n]$, and let P be their combined oracle. There is a realization for which, with $H = X_{[n]}$,

$$P <_T H <_T P', \quad H' \equiv_T P'. \tag{1.2}$$

In particular each $X_i \leq_T P'$.

The assumptions place no further restriction on how much information can appear when participants cooperate. The ideals for overlapping coalitions need not obey an intersection identity, and the ideal assigned to a union may be strictly larger than the ideal generated by the two constituent ideals. The theorem asserts a complete finite classification at the level of *cores*, not of the full degree spectra or of coarse reducibility.

Corollary 1.2 (Arbitrary two-oracle synergy). *For every countable Turing ideal $\mathcal{I}$, there are X, Y such that*

$$\text{Core}(X) = \text{Core}(Y) = \text{Comp}, \quad \text{Core}(X \oplus Y) = \mathcal{I}.$$

They can be chosen individually 1-generic, with no least representative in any of the three relevant classes of X , Y , or $X \oplus Y$.

A direct two-oracle construction, stronger than merely applying the main theorem, appears in Section 6. The main theorem itself is proved in Sections 7–9. The construction is finite in its sharing architecture but infinite in its oracle coding and genericity.

1.2 Why the upper bound is the real issue

To make every authorized coalition recover a chosen ideal, it is enough to supply redundant encodings of its generators. That establishes only one inclusion in (1.1). It does not prevent masks, combinations of shares, or the choice of the presentation from encoding additional robust information. Finite probabilistic secrecy by itself also does not establish an equality of Turing ideals.

Our proof handles this through three levels. The code for an ideal can be approximated in density by codes computable from finitely many generators. For a fixed coalition, every partly observed sharing component can be absorbed into a change of generic coordinates. Finally, a cone-avoidance compactness theorem converts information surviving all sparse errors into information surviving a whole positive-radius neighborhood. This lets us replace each full secret code by a finite approximation and eliminate the generic coordinates. The surviving information must belong to the prescribed ideal.

2 Repository provenance and dependency boundary

2.1 What is already in ProveIt

The repository was inspected at commit

1a1396d4d3a2ac6812692df520517d86ba3a4785.

The relevant program is `Computability/TuringDegrees/Research/CoarseDegrees/`. Its research plan already states target C4 and gives the precise dyadic column construction for realizing one countable Turing ideal as a core [4]. We verify that sketch in Section 5; we do not claim it as a new discovery. The same column-coding architecture occurs in the proof of Theorem 3.11(2) of Hirschfeldt–Jockusch–Schupp [2], together with their coarse-description projection lemma.

The repository’s synthesis already develops equality of representative spectra, criteria for a least representative, and examples with prescribed unattained principal infima [5]. The no-least statement below is therefore not advertised in isolation as a new phenomenon. Its role here is that it can be imposed *simultaneously* on every nonempty coalition while its entire ideal-valued profile is prescribed.

The finite access-structure architecture is related to the classical construction of general secret-sharing schemes by Ito, Saito, and Nishizeki [3]. We prove the specific additive algebra needed here directly. No novelty is claimed for splitting one bit into exclusive-or shares. The proposed contribution is the combination with exact coarse-core upper bounds, arbitrary countable ideals, and a common relative jump bound.

2.2 Claim ledger

Statement	Role in this article	Status / attribution
Cone-avoiding compactness; generic trivial core	Two non-elementary inputs	Published, [1]
Single-ideal column code	Complete verification of C4	Already sketched in [4]; coding in [2]
Finite affine sharing	Algebraic mechanism	Classical architecture; direct proof supplied
All finite monotone profiles	Main classification theorem	Proposed contribution; full conventional proof
Simultaneous nonattainment and relative lowness	Strengthening of realization	Proved here for the constructed profile
Unauthorized coalitions generic over the secret presentation	Access-structure refinement	Proved here
Uniform exact core is computable	Clarification of C4's uniformity question	Elementary established obstruction, not a novelty claim

A search and source comparison did not establish priority for the main combined statement. The inspected sources and the repository's stated results do not supply that statement in this form. This is a bounded literature assessment, not evidence that no equivalent theorem exists. Independent mathematical review and a broader priority search are still appropriate before publication.

2.3 Formal verification is a separate claim

The repository distinguishes its genuine oracle-recursion development from a coarse-degree library containing explicitly admitted literature inputs [6]. This article does not rely on an uninspected assertion that the relevant Lean declarations are assumption-free. Its proofs are ordinary mathematical proofs with the two published inputs isolated below. No Lean or Rocq build was performed for this article. The accompanying Python program checks finite identities only; its results are not a substitute for a proof of an infinite computability theorem.

2.4 [write] Notation against the research plan and the other round-2 reports

The notation of this article (Appendix A) agrees with the programme's research plan [4] and synthesis [5] in the objects they share: Core, Low, Comp, the Turing spectra of coarse classes, and the block code J of (5.1), which is the plan's J (constant on $[2^n, 2^{n+1})$) and the Lean library's `blockCode`, including the value $J(A)(0) = 0$. No symbol was renamed and no normalization was changed when this report was written. The readings most likely to mislead a reader who comes from the plan, the synthesis, the Lean library or the neighbouring reports:

Symbol	Meaning here	Do not read it as
R_k, r_k	the k th dyadic column $\{x : v_2(x+1) = k\}$, $r_k(t) = 2^k(2t+1) - 1$; the fibre over k of the Lean column index <code>col</code>	the dyadic <i>code</i> $R(A)(n) = A(v_2(n+1))$ of the plan and synthesis (Lean <code>Rc</code>); this article never uses that code
$R, R_{S,j}$	a generic real (Theorem 6.1); the mask streams of an all-of- S component	a column, or the dyadic code
Z, Z_S	the ideal code (5.4); it is the set X of the plan's proposed C4 proposition	a presentation oracle, which is P here
$Z^{[m]}$	the first- m -column truncation, computable in B_m	(it is the plan's Y_k with $k = m$)
P	an oracle presenting all generators at once	a member of the ideal: it need not be one
X_C	the specified periodic interleaving of the X_i , $i \in C$	an arbitrary representative of the join degree
$X \triangle Y, X \oplus Y$	symmetric difference; oracle join	each other
$D \approx X$	$\delta(D, X) = 0$, i.e. the Lean <code>SetCoarseEq D X</code> (density-zero symmetric difference)	closeness at some positive radius
$\text{Spec}_{\text{nc}}, \text{Spec}_{\text{uc}}$	Turing spectra of coarse classes (Lemma 9.1)	coarse degrees; or the <i>effective-dense</i> spectra Spec_T of report 14
1-generic relative to B	Definition 3.5; the Lean predicate <code>OneGenericRel B G</code>	1-generic and merely Turing-above B

The other reports of round 2 name their objects independently. Reports 12 and 13 attach letters such as X_A , F and J to codes with other block conventions (for example blocks $[2^n - 1, 2^{n+1} - 1]$), and report 14's G_T is an effective-dense object, not a coarse one. Statements should be compared through their definitions, not through their letters.

2.5 [write] Provenance of this report

The report is built from one manuscript: batch 37 manuscript 07, the archive `Coarse_Information_Profiles` (inner directory `coarse_information_profiles/`, main file `article.tex`, 28-page PDF), which arrived in commit 37e61c1fd. Its pin is 1a1396d4d3a2ac6812692df520517d86ba3a4785, the commit displayed in Section 2 (the batch-36 placement commit). It was placed as report 11 in commit 0e53d1034. Between the pin and the placement, `Computability/TuringDegrees/` changed only by the addition of the four round-2 reports 11–14, so every repository statement in Section 2 and Appendix B is true of the files it names in the current tree. What the manuscript could not know is the rest of its round: reports 12 and 13 also prove C4, and report 14 takes up C2.

When the report was written, every label received the prefix `cr11:`, and the title-page note, these four subsections, and [write] notes after Theorem 5.2, in Section 12.2 and in Appendix B were added. No statement of the manuscript was changed. Every numbered statement, equation and section keeps the number it has in the delivered PDF; that PDF is not shipped, and `article.pdf` is a build of this text.

The delivered checks are shipped under new paths: `checks/verify_finite.py` is `code/verify_finite.py`, `checks/results.json` is `data/results.json`, `validation.json` is `data/validation.json`, and `build.sh` is `code/build.sh`; `PROOF_AUDIT.md` and `SOURCES.md` keep their names. The delivered checksum file is not shipped. The delivered `PROOF_AUDIT.md` names the two published inputs “Theorems 3.3 and 3.5”; in this article they are Theorem 3.3 and Theorem 3.6, since Definition 3.5 shares the theorem counter. The audit is shipped unchanged.

2.6 [write] Formal status and the Lean library

No statement of this report is formalized, in Lean or in Rocq. That the report sits in the research programme of the Lean/Rocq project Computability/TuringDegrees gives it no formal status. The project’s Lean library CoarseDegrees (Computability/TuringDegrees/Lean/CoarseDegrees), which closes eight published statements with `admit`, touches the report as follows.

- *Relied on, admitted there too.* Theorem 3.6, the relativized HJKS Theorem 4.2, is the admitted `OneGenericRel.core_le` (Cone.lean, line 221): if $A \leq_T B \oplus D$ for every $D \approx G$, then $A \leq_T B$. With computable background, as in Theorem 6.1 and Theorem 10.2, it is the admitted `OneGeneric.core_trivial` (Published.lean, line 48). Theorem 3.3 (HJKS Theorem 3.7) and Corollary 3.4 have no Lean statement.
- *Proved here informally, admitted in Lean.* Lemma 4.2 proves, in the stronger form $\delta(G, E) = 1$, the relativized Jockusch–Schupp statement that the library admits as `OneGenericRel.not_coarselyComputableIn` (Cone.lean, line 228): a set 1-generic relative to P has no P -computable coarse description. The unrelativized case is proved in Lean as `OneGeneric.not_setCoarselyComputable` (GenericDensity.lean, line 124). The proof given here is not formalized, so the Lean admission stands.
- *Proved in Lean, with nothing admitted, and reproved here.* Lemma 5.1 is `blockCode_decode` (Block.lean, line 254) for the same code; the tail count (5.3) is `card_tail` (Columns.lean, line 74); the upward closure in Lemma 9.1 is `exists_description_of_degree` (BlockChar.lean, line 42), which implants on $\{2^k - 1\}$ where the article uses $\{2^k\}$; the existence of relatively 1-generic sets is `exists_oneGenericRel` (Cone.lean, line 196), without the bound $G \leq_T P'$ of Lemma 4.4, since the library has no jump operator. Proposition 9.2 is the core-member clause of the synthesis’s least-representative theorem; the library proves that theorem’s block-code clause, `isLeastNC_iff_blockCode` (BlockChar.lean, line 67), not this clause.
- *Not formalized.* Everything else, in particular Theorem 1.1, the C4 identity of Theorem 5.2, Proposition 5.3, Theorem 6.1, Sections 7–11 and the jump calculations. Section 12.3 is a plan: no Lean or Rocq code was delivered or is shipped. The finite checks are ordinary Python.

2.7 [write] Programme targets addressed, and what stays open

Line numbers refer to Computability/TuringDegrees/Research/CoarseDegrees/research-plan/turing_degrees_unified.tex (“plan”) and .../research-synthesis/Turing_Degrees_Synthesis.tex (“synthesis”), which are identical at the pin and in the current tree.

- C4 (plan:585–598, “V: proof sketch supplied; novelty unchecked”). Theorem 5.2 is the plan’s proposed Proposition (plan:588) with the same columns and the same J , and its proof is the plan’s sketch written out, with the robust radius taken from HJKS Theorem 3.7 as in the plan’s (K3). Of the four tasks listed at plan:598: the sketch is checked (by this unrefered text); the literature search found the column-coding architecture in HJS 2021, Lemma 3.10 and Theorem 3.11(2) (Section 2, SOURCES.md), so “novelty unchecked” becomes “close published antecedent”; the question for which sequences (A_k) the class has a least degree is not answered here beyond the general criterion of Proposition 9.2 (report 12, directory ../12, takes it up for its code); and the uniform core is settled only for literal exact recovery, by Proposition 5.3, which says nothing about uniform coarse reducibility (C8). The quantitative exact-pair consequence proposed at plan:598 is not addressed.

Plan:1647 lists C4 among the quickest available gains and asks the next round to spread over C2 and C4–C7; reports 12 and 13 are the other C4 attempts of the round.

- *Classification* (synthesis:973). Its first half, which countable ideals occur as cores, is answered: all of them (Theorem 5.2, and reports 12 and 13 independently). The second half, which upward-closed sets of degrees occur as spectra, stays open: this report computes cores, not spectra (Section 5.2, Question 13.5).
- *Minimal elements* (synthesis:970) and C5 (plan:600): not settled (end of Section 9.2, Question 13.5).
- C2 (plan:563, synthesis:969) and C8 (plan:609, synthesis:971): not addressed; Question 13.7 records that the density-zero argument does not transfer to effective-dense descriptions. Report 14 is the round's C2 attempt.
- *Beyond the plan*. Theorem 1.1, Theorem 6.1, Theorem 10.2 and Proposition 11.1 answer no plan target or synthesis item; they are the manuscript's proposed contribution, and their priority is not established.

3 Basic density and oracle calculus

3.1 Periodic coordinates and finite operations

For $m \geq 1$ and $Y_0, \dots, Y_{m-1} \in 2^{\mathbb{N}}$, write

$$\text{Join}_m(Y_0, \dots, Y_{m-1})(mt + j) = Y_j(t) \quad (0 \leq j < m).$$

There is a computable coordinate extraction map for every j . For sets E_j on the coordinates, counting at lengths divisible by m gives

$$\rho_{mN}(\text{Join}_m(E_0, \dots, E_{m-1})) = \frac{1}{m} \sum_{j < m} \rho_N(E_j). \quad (3.1)$$

At other lengths the difference is bounded by a term tending to zero.

Lemma 3.1 (Density calculus). *The following facts hold.*

- (a) *If $D \approx \text{Join}_m(Y_j)$, then every extracted coordinate D_j satisfies $D_j \approx Y_j$.*
- (b) *If $D_j \approx Y_j$ for all $j < m$, then $\text{Join}_m(D_j) \approx \text{Join}_m(Y_j)$.*
- (c) *For two tuples,*

$$\delta(\text{Join}_m(Y_j), \text{Join}_m(Z_j)) \leq \frac{1}{m} \sum_{j < m} \delta(Y_j, Z_j).$$

- (d) *For finite exclusive-or sums,*

$$\delta\left(\bigtriangleup_{j < r} Y_j, \bigtriangleup_{j < r} Z_j\right) \leq \sum_{j < r} \delta(Y_j, Z_j).$$

Consequently, finite coordinate projections, permutations, duplications, and pointwise exclusive-or operations preserve coarse similarity whenever all their input coordinates are coarsely similar.

Proof. For (a), an error in coordinate j before its N th position is an error in the full oracle before position mN . Hence its error fraction is at most $m\rho_{mN}(D \triangle \text{Join}_m(Y_j))$, which tends to zero. Parts (b) and (c) follow from (3.1) and finite subadditivity of the limsup. For (d), if the two exclusive-or outputs differ at a position, at least one corresponding pair of inputs differs there. Thus the output error set is contained in the finite union of the input error sets. The last assertion follows by composition of these operations. A fixed extra oracle used identically on both sides introduces no errors. $\square$

A permutation of finitely many periodic coordinates preserves δ exactly: at lengths mN it preserves the error count. Regrouping a finite join of equally sized tuples into one tuple has the same effect. Later each participant receives exactly the same number of streams, so flattening coalition joins has no hidden density distortion.

Lemma 3.2 (Necessary profile conditions). *For every X , $\text{Core}(X)$ is a countable Turing ideal. If $C \subseteq D$, then $\text{Core}(X_C) \subseteq \text{Core}(X_D)$. Also $\text{Core}(X_\emptyset) = \text{Comp}$.*

Proof. Every lower cone is downward closed and closed under finite joins. The core is an intersection of such cones and contains all computable sets. Since X itself is a coarse description, $\text{Core}(X) \subseteq \text{Low}(X)$; the latter is countable because there are countably many oracle programs. Given a coarse description of X_D , extract and regroup the coordinates belonging to C . Lemma 3.1 produces a coarse description of X_C , proving monotonicity. The empty coalition is represented by a computable oracle, which proves the final assertion. $\square$

3.2 Relative cores and the two published inputs

For a background oracle B , define

$$\text{Core}^B(X) = \{A : (\forall D \approx X) A \leq_T B \oplus D\}.$$

All descriptions here are arbitrary binary sets; they are not required to be computable in the background oracle.

Theorem 3.3 (Cone-avoiding compactness, published input). *If for every $\varepsilon > 0$ there is a Y with $\delta(X, Y) \leq \varepsilon$ and $A \not\leq_T B \oplus Y$, then there is a $D \approx X$ with $A \not\leq_T B \oplus D$.*

This is Theorem 3.7 of [1], with all oracle computations relativized to B . Its contrapositive supplies the form we use.

Corollary 3.4 (Robust radius). *For all A, B, X ,*

$$A \in \text{Core}^B(X) \iff (\exists \eta > 0)(\forall Y)[\delta(X, Y) < \eta \implies A \leq_T B \oplus Y]. \quad (3.2)$$

Proof. If no such radius exists, for every positive ε choose a counterexample at distance less than ε and apply Theorem 3.3. Conversely, every coarse description has distance zero and hence is within any positive radius. The strict inequality avoids irrelevant boundary conventions; replacing a radius by a smaller one would give the same use. $\square$

Definition 3.5. A sequence G is 1-generic relative to B if for every B -computably enumerable set W of finite binary strings, either some prefix of G belongs to W , or some prefix of G has no extension in W .

Theorem 3.6 (Generic core, published input). *If G is 1-generic relative to B , then*

$$\text{Core}^B(G) = \text{Low}(B). \quad (3.3)$$

This is the relativization of Theorem 4.2 of [1]; the background oracle is present in every recovery computation. We use this precise relative formulation, not an inference that ordinary genericity is enough in the presence of an arbitrary background oracle. These two named inputs are the only non-elementary coarse-computability theorems used below.

4 Generic coordinates and a transfer principle

4.1 The elementary genericity facts needed later

We record the details to identify precisely which genericity properties the construction uses. Finite tuples of binary streams are always identified with one binary sequence by periodic interleaving.

Lemma 4.1 (Changes and projections of generic coordinates). *Let G be 1-generic relative to P .*

- (a) *If $B \leq_T P$, then G is 1-generic relative to B .*
- (b) *If h is a P -computable homeomorphism of a finite product of Cantor spaces with P -computable inverse, then $h(G)$ is 1-generic relative to P .*
- (c) *Every nonempty finite coordinate projection of G is 1-generic relative to P .*

In particular, invertible affine changes over $\mathbb{F}_2$, applied pointwise to finitely many streams with P -computable offsets, preserve 1-genericity relative to P .

Proof. Part (a) follows because every B -c.e. set of strings is P -c.e. For (b), view a c.e. set of strings as the effectively open union of its cylinders. The inverse image under h is P -effectively open. If G belongs to it, then $h(G)$ meets the original open set. Otherwise genericity gives a cylinder about G disjoint from the inverse image. Its image is an open neighborhood of $h(G)$ disjoint from the original open set; it contains a cylinder about $h(G)$, as required.

For (c), pull back the effectively open set by the coordinate projection. If a cylinder about G avoids this pullback, its projection is an open neighborhood of the projected sequence and avoids the original set. The projection of a basic finite-coordinate cylinder is again open. This proves the avoiding alternative. The meeting alternative is immediate. Finally, the stated affine maps and their inverses are pointwise finite Boolean operations, so they satisfy (b). $\square$

Lemma 4.2 (Genericity excludes a background-computable description). *If G is 1-generic relative to P and $E \leq_T P$, then*

$$\delta(G, E) = 1.$$

In particular, G has no P -computable coarse description.

Proof. Fix a rational $q < 1$ and an integer b . Consider strings σ having a length $N > b$ with $\rho_N(\sigma \Delta E) > q$. They form a P -c.e. upward-closed family, where membership is witnessed by a finite prefix. It is dense: extend any string by a sufficiently long interval on which the bits are the complements of the corresponding bits of E . A generic sequence must meet a dense c.e. family, since the avoiding alternative is impossible. Thus there are arbitrarily large such N for every rational $q < 1$. The limsup is one. $\square$

Lemma 4.3 (No noncomputable base). *If G is 1-generic relative to P , $A \leq_T P$, and $A \leq_T G$, then A is computable.*

Proof. Choose Φ with $\Phi^G = A$. The set of strings σ for which $\Phi^\sigma(t) \downarrow \neq A(t)$ for some t is P -c.e. No prefix of G belongs to it. Hence a prefix τ of G has no extension in this set. To compute $A(t)$ without an oracle, search finite extensions of this fixed τ for any convergent computation $\Phi^\sigma(t)$. Such an extension exists because Φ^G is total. Its output must equal $A(t)$ by the choice of τ . Hard-coding the finite string τ gives an ordinary program for A . $\square$

Lemma 4.4 (A bounded generic with a controlled jump). *For every P there is a $G \leq_T P'$ that is 1-generic relative to P . Every such G satisfies*

$$(P \oplus G)' \equiv_T G \oplus P' \equiv_T P'. \quad (4.1)$$

Consequently $P <_T P \oplus G <_T P'$.

Proof. List the P -c.e. families W_e of finite strings. Starting with an empty string σ_e , use P' to decide whether there is an extension in W_e . If one exists, find one by P -effective search and extend to it; if none exists, leave the current string unchanged at this substep. Append a bit to ensure unbounded length. The union G is P' -computable and meets or avoids each W_e .

For the jump assertion, fix an index e for a computation with oracle $P \oplus G$. The family of finite G -strings on which this computation halts is P -c.e. To decide its outcome using $G \oplus P'$, search prefixes σ of G until either the computation already halts on σ , or P' reports that no extension of σ permits it to halt. Genericity guarantees that one alternative eventually occurs. Thus $(P \oplus G)' \leq_T G \oplus P'$. The reverse reduction uses $G \leq_T (P \oplus G)'$ and monotonicity of the jump. Since $G \leq_T P'$, the last degree is P' .

Lemma 4.2 implies $G \not\leq_T P$, giving the first strict inequality. If $P \oplus G \equiv_T P'$, jumping and using (4.1) would give $P'' \equiv_T P'$, contrary to strictness of the Turing jump. $\square$

The bound is relative to the *chosen presentation oracle*. A countable ideal may have many presentations of different degrees. No canonical least presentation is assumed, and no absolute lowness conclusion is asserted unless P itself is computable.

4.2 Generic perturbations of successively simpler data

The following lemma packages the upper-bound argument. Its strength comes from quantifying over *every* coarse description of a generic oracle, not merely the chosen exact generic sequence.

Lemma 4.5 (Generic-perturbation transfer). *Suppose W is a binary sequence and, for each m , there are:*

- (i) *an oracle B_m and a sequence G_m that is 1-generic relative to B_m ;*
- (ii) *a total B_m -computable oracle map $F_m : 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ such that $D \approx G_m$ implies $F_m(D) \approx F_m(G_m)$;*
- (iii) *a bound $\varepsilon_m \rightarrow 0$ with $\delta(W, F_m(G_m)) \leq \varepsilon_m$.*

Then each $A \in \text{Core}(W)$ satisfies $A \leq_T B_m$ for every sufficiently large m . In particular, if the lower cones $\text{Low}(B_m)$ are increasing,

$$\text{Core}(W) \subseteq \bigcup_m \text{Low}(B_m). \quad (4.2)$$

Proof. Let $A \in \text{Core}(W)$ and take a radius $\eta > 0$ from Corollary 3.4, with computable background. For every large m we have $\varepsilon_m < \eta$. Fix such an m and any $D \approx G_m$. The density triangle inequality gives

$$\delta(W, F_m(D)) \leq \delta(W, F_m(G_m)) + \delta(F_m(G_m), F_m(D)) \leq \varepsilon_m < \eta.$$

Therefore $A \leq_T F_m(D) \leq_T B_m \oplus D$. As D was arbitrary, $A \in \text{Core}^{B_m}(G_m)$. Theorem 3.6 gives $A \leq_T B_m$. The same argument works for every sufficiently large m . If the lower cones are increasing, their eventual-membership union is simply the union in (4.2). $\square$

There is no assumption that A is computable from the master presentation used to construct the objects. That fact is obtained as a conclusion when the B_m belong to the intended ideal. This prevents a circular cone-avoidance argument in which the object to be excluded is silently placed into the background oracle.

5 Verifying C4: one countable ideal as a core

5.1 Robust blocks and positive-density columns

For $A \in 2^{\mathbb{N}}$, define

$$J(A)(0) = 0, \quad J(A)(t) = A(j) \quad \text{if } 2^j \leq t < 2^{j+1}. \quad (5.1)$$

The map is uniform in A , and $J(A) \equiv_T A$ by reading the positions 2^j .

Lemma 5.1 (Nonuniform block decoding). *Every coarse description D of $J(A)$ computes A .*

Proof. On the block $[2^j, 2^{j+1})$, compute the majority value of D , assigning zero in a tie. If this value is wrong, at least 2^{j-1} positions of the block are errors. Thus the error density at the endpoint 2^{j+1} is at least $1/4$. This can happen for only finitely many j when $D \approx J(A)$. The majority sequence, computable from D , is therefore a finite variant of A . Hard-code its finitely many exceptions to obtain a reduction of A to D . $\square$

The finite correction depends on D . We have proved $\forall D \exists e$, not $\exists e \forall D$. Neither a bound on the number of exceptional blocks nor a procedure locating them is part of the conclusion.

Partition $\mathbb{N}$ into the computable columns

$$R_k = \{r_k(t) : t \in \mathbb{N}\}, \quad r_k(t) = 2^k(2t + 1) - 1. \quad (5.2)$$

The partition is disjoint and exhaustive because k is the exponent of two in $x + 1$. The column R_k has density $2^{-(k+1)}$. More precisely,

$$\left| [0, N) \cap \bigcup_{k \geq m} R_k \right| = \left\lfloor \frac{N}{2^m} \right\rfloor. \quad (5.3)$$

Indeed, membership in this union means that $x + 1$ is divisible by 2^m .

Given a sequence (A_k) , define its ideal code Z by

$$Z(r_k(t)) = J(A_k)(t). \quad (5.4)$$

Let $Z^{[m]}$ agree with Z on $R_0, \dots, R_{m-1}$ and be zero on every other column. Then

$$Z^{[m]} \leq_T B_m := \bigoplus_{k < m} A_k, \quad \delta(Z, Z^{[m]}) \leq 2^{-m}. \quad (5.5)$$

The empty join B_0 is computable.

Theorem 5.2 (C4, with the sketch fully verified). *The code in (5.4) satisfies*

$$\text{Core}(Z) = \bigcup_m \text{Low}(B_m). \quad (5.6)$$

Consequently every countable Turing ideal is the core of a binary sequence. Moreover, the exact code Z is Turing equivalent, uniformly, to the full presentation $P_A = \bigoplus_{k \in \mathbb{N}} A_k$.

Proof. Let $D \approx Z$ and write $E = D \triangle Z$. Extract the k th column as $D_k(t) = D(r_k(t))$. Its first N positions lie below $2^{k+1}N$, so

$$\rho_N(D_k \triangle J(A_k)) \leq 2^{k+1} \rho_{2^{k+1}N}(E) \longrightarrow 0.$$

Thus D_k is a coarse description of $J(A_k)$ and computes A_k by Lemma 5.1. A finite join of these reductions shows $B_m \leq_T D$ for every m . This proves the inclusion from right to left in (5.6).

For the reverse inclusion, let $A \in \text{Core}(Z)$ and choose a robust radius η from Corollary 3.4. Choose m with $2^{-m} < \eta$. Then (5.5) gives $A \leq_T Z^{[m]} \leq_T B_m$. This proves the equality. If $\mathcal{I}$ is a countable ideal, enumerate a generating sequence from it. Its closure under finite joins and downward reducibility is exactly the right side of the equality.

Finally, P_A computes Z directly from (5.4). Conversely, to obtain $A_k(j)$ from the exact code, read $Z(r_k(2^j))$. This recovers the two-parameter array uniformly, proving the last assertion. $\square$

[write] Theorem 5.2 is the proposed Proposition of target C4 in the research plan (turing_degrees_unified.tex, line 588), with the same columns, the same block code, and Z in the role of the plan's X ; the proof above is the plan's sketch with every step written out. Reports 12 and 13 of the same round (directories ../12 and ../13) prove the same identity for their own codes, report 12 in stronger forms. None of these proofs is formalized (Section 2.6).

5.2 Two quantifier distinctions

Every coarse description of Z computes each generator A_k separately. It need not compute their infinite join P_A . The reductions to A_k may require unrelated finite corrections, with no computable procedure choosing the corrections as a function of k . The exact equivalence $Z \equiv_T P_A$ is therefore consistent with a nonprincipal core strictly below $\text{Low}(P_A)$.

The theorem also does not say that every degree which is an upper bound for all the A_k computes a coarse description of Z . An arbitrary family of nonuniform upper-bound reductions need not yield a usable approximation to the full column array. Thus the theorem is an exact *core* calculation, not an asserted complete spectrum calculation.

5.3 Uniform exact recovery collapses

One question attached to C4 concerns replacing its nonuniform quantifier by a single recovering functional. In that literal exact-output sense the answer is completely different, for a reason unrelated to the particular column code.

Proposition 5.3 (Uniform exact core). *For every X, B , the collection of sets A for which some fixed functional Φ satisfies*

$$(\forall D \approx X) \quad \Phi^{B \oplus D} = A$$

is exactly $\text{Low}(B)$. In particular the unrelativized uniform exact core of every X is Comp .

Proof. Suppose such a Φ exists. For a given input t , search with oracle B through all finite binary strings σ and all finite computations until $\Phi^{B \oplus \sigma}(t)$ converges. This search terminates since $\Phi^{B \oplus X}(t)$ does. Every finite σ has an extension D that is a finite variant of X : replace the corresponding prefix of X by σ . Hence $D \approx X$, and the convergent finite computation must have output $A(t)$. The search computes A in B . Conversely, when $A \leq_T B$, simply ignore the D -coordinate. $\square$

This is an exact-recovery obstruction, also reflected in the discussion of uniform decoding in [1]. It is not a statement that all *uniform coarse reductions* are trivial: such a reduction only has to output some coarse description of its target, and different input oracles may produce different output sequences.

6 Two masks reveal an arbitrary ideal

The smallest instance already captures the reason why cores do not determine how information combines.

Theorem 6.1 (Two-share realization). *Let $\mathcal{I}$ be a countable Turing ideal and let P present a generating sequence. There are binary X, Y such that*

$$\text{Core}(X) = \text{Core}(Y) = \text{Comp}, \quad \text{Core}(X \oplus Y) = \mathcal{I}. \quad (6.1)$$

Both X and Y are 1-generic relative to P . They and their join have no P -computable coarse descriptions. With a suitable choice, their full join $H = X \oplus Y$ satisfies (1.2).

Proof. Let Z be the code of Theorem 5.2, so $Z \equiv_T P$. Choose R 1-generic relative to P , and put

$$X = R, \quad Y = R \triangle Z.$$

Translation by the P -computable sequence Z is an invertible P -computable homeomorphism. Thus both sequences are 1-generic relative to P , hence also without an oracle. Theorem 3.6 with computable background gives their two trivial cores.

If $D \approx X \oplus Y$, its two coordinates D_0, D_1 satisfy $D_0 \approx R$ and $D_1 \approx R \triangle Z$. Therefore $D_0 \triangle D_1 \approx Z$. Every member of $\mathcal{I}$ is computed by this coarse description of Z , giving $\mathcal{I} \subseteq \text{Core}(X \oplus Y)$.

For the opposite inclusion, set

$$F_m(V) = V \oplus (V \triangle Z^{[m]}), \quad B_m = \bigoplus_{k < m} A_k.$$

This map is B_m -computable and preserves coarse similarity by Lemma 3.1. Since only the second coordinate changes,

$$\delta(X \oplus Y, F_m(R)) \leq 2^{-m-1}.$$

Also R is 1-generic relative to every $B_m \leq_T P$. Lemma 4.5 now gives $\text{Core}(X \oplus Y) \subseteq \bigcup_m \text{Low}(B_m) = \mathcal{I}$.

Neither X nor Y has a P -computable coarse description by Lemma 4.2. Such a description of their join would project to one of X , so it too is impossible. Finally, $H \equiv_T R \oplus Z \equiv_T R \oplus P$. Choosing $R \leq_T P'$ as in Lemma 4.4 gives the jump bound and the strict inequalities. $\square$

The assertion about no least representatives will follow from the general spectrum criterion in Section 9. Every member of the three cores just calculated is computable in P , whereas none of these targets has a P -computable description.

Corollary 6.2 (There is no core-only operation for oracle join). *There is no operation $\mathfrak{F}$ on pairs of countable Turing ideals satisfying*

$$\text{Core}(X \oplus Y) = \mathfrak{F}(\text{Core}(X), \text{Core}(Y)) \quad \text{for all } X, Y.$$

Proof. The pair of inputs (Comp, Comp) can have every countable Turing ideal as its output by Theorem 6.1, including two different ideals. $\square$

Example 6.3 (A principal secret and a nonprincipal secret). For a noncomputable A , take a presentation of $\mathcal{I} = \text{Low}(A)$. The individual shares have trivial cores, while their join robustly computes exactly the sets below A . Its core has greatest degree $\deg_T(A)$, but that degree will not be the degree of any coarse representative.

For a nonprincipal example, let $\mathcal{I}$ be the ideal generated by the finite Turing jumps $\emptyset^{(k)}$, $k \in \mathbb{N}$. The strictness of the jump makes this ideal nonprincipal. Each finite jump survives in every coarse description of the joined shares, but their infinite presentation need not survive. In this case the representative spectrum has no greatest lower bound at all; see Corollary 9.5.

7 Finite additive sharing and coalition normal forms

7.1 An all-of- S component

Let $S = \{s_1 < \dots < s_r\}$ be a nonempty set of participants, and let Z_S be a binary secret stream. For $r \geq 2$, introduce $r - 1$ mask streams $R_{S,1}, \dots, R_{S,r-1}$. Give participant s_j the stream

$$W_{S,s_j} = \begin{cases} R_{S,j}, & j < r, \\ Z_S \Delta R_{S,1} \Delta \dots \Delta R_{S,r-1}, & j = r. \end{cases} \quad (7.1)$$

For $r = 1$, give its unique participant $W_{S,s_1} = Z_S$. Always,

$$\bigtriangleup_{i \in S} W_{S,i} = Z_S. \quad (7.2)$$

This is the usual additive all-of- r sharing identity. The point of the next lemma is not that the shares have a probability distribution: no randomness is assumed. The lemma identifies the observations as a projection of a computable change of generic coordinates.

Lemma 7.1 (Proper observations are free coordinates). *Suppose $Z_S \leq_T P$. Let O be a proper subset of the r share positions in (7.1). The tuple of shares observed in O is a coordinate projection of a P -computable invertible affine change of the $r - 1$ masks. If O is empty this is the empty projection. If O is nonempty and the mask tuple is 1-generic relative to P , the observed tuple is 1-generic relative to P .*

Proof. If the last share is not observed, the observations are already distinct raw masks. Otherwise the last share is observed but at least one earlier mask position, say j , is missing. Replace the j th mask coordinate by

$$\widetilde{R}_{S,j} = Z_S \Delta \bigtriangleup_{\ell < r} R_{S,\ell}, \quad \widetilde{R}_{S,\ell} = R_{S,\ell} \quad (\ell \neq j). \quad (7.3)$$

This is invertible: the inverse recovers the omitted mask by

$$R_{S,j} = Z_S \Delta \widetilde{R}_{S,j} \Delta \bigtriangleup_{\ell < r, \ell \neq j} \widetilde{R}_{S,\ell}.$$

The new j th coordinate is precisely the observed last share; all observed earlier shares are unchanged. Distinct observations therefore become distinct new mask coordinates. Since $Z_S \leq_T P$, the map and its inverse are P -computable. Apply Lemma 4.1 for the last assertion. When $r = 1$, a proper observation can only be empty, which has already been excluded from the genericity assertion. $\square$

7.2 Assembling the complete finite profile

Fix data satisfying Theorem 1.1. For each nonempty $S \subseteq [n]$, choose generators $(A_{S,k})_{k \in \mathbb{N}}$ of $\mathcal{I}_S$ and let Z_S be their ideal code. Fix a computable pairing convention and put

$$P = \bigoplus_{\substack{\emptyset \neq S \subseteq [n] \\ k \in \mathbb{N}}} A_{S,k}. \quad (7.4)$$

In particular $Z_S \leq_T P$ uniformly in the finite parameter S .

Introduce disjoint coordinates for every mask in every all-of- S component. In addition, introduce one *private generic stream* T_i for each participant i . Let G be the regular finite join of all these mask and private coordinates, chosen 1-generic relative to P . The number of its coordinates is

$$Q = n + \sum_{\emptyset \neq S \subseteq [n]} (|S| - 1) = n + (n - 2)2^{n-1} + 1 \geq 1. \quad (7.5)$$

The streams are projections of one generic tuple. We do not separately choose generic masks and then assume that their joint tuple is generic.

Participant i receives its private stream T_i and its share $W_{S,i}$ for every S containing i . There are exactly

$$L = 2^{n-1} + 1 \quad (7.6)$$

streams per participant. Define X_i to be their regular L -way join in a fixed computable order. For a coalition C , flatten its join to the $M_C = |C|L$ observed streams. This differs from any other fixed ordering only by a finite periodic-coordinate permutation.

The private streams are not needed for the lower inclusion of cores. Their purpose is to guarantee that every nonempty coalition contains an unremovable generic obstruction to a P -computable coarse description. They will also ensure that the normal form always has at least one generic coordinate.

Lemma 7.2 (Coalition normal form). *For every nonempty $C \subseteq [n]$, there is a finite tuple V_C which is 1-generic relative to P , and a fixed pointwise finite exclusive-or and interleaving map F_C , such that*

$$X_C = F_C(V_C, (Z_S)_{\emptyset \neq S \subseteq C}). \quad (7.7)$$

Here equality uses the fixed flattened coordinate ordering. The map F_C is an ordinary computable map when the displayed secrets are supplied as additional inputs. No secret Z_S with $S \not\subseteq C$ is an additional input. The number of free stream coordinates in V_C is

$$q_C = |C|L - (2^{|C|} - 1) \geq |C|. \quad (7.8)$$

Proof. Examine the components separately.

If $S \subseteq C$, all its shares are observed. Keep its $|S| - 1$ raw masks as free coordinates. Reconstruct its last share using those masks and Z_S . When $|S| = 1$, there is no free coordinate and the observed stream is just Z_S .

If $\emptyset \neq S \cap C \neq S$, apply Lemma 7.1 to the proper set of observed positions. All observed shares become free coordinates of an invertible P -computable affine change of the component's masks. Their reconstruction needs no additional secret input: in the new coordinates the observed streams are copied verbatim. The unauthorized secret may enter the *change of coordinates*, but after that change it is not an input to the reconstruction map.

Components with $S \cap C = \emptyset$ make no observations. Private streams T_i for $i \in C$ are kept as free coordinates. The affine changes for the different components act on disjoint groups of coordinates, so together with the identity on private streams they form one P -computable homeomorphism of the master product space. Project its output to all the chosen free coordinates. Lemma 4.1 proves that the result V_C is 1-generic relative to P . It is a nonempty projection because C contains at least one private stream.

There is one parity constraint for each fully observed component, and no constraint for a proper observation. There are $2^{|C|} - 1$ nonempty subsets of C , so the number of free coordinates is the total number $|C|L$ of observed streams minus this number of constraints. Each constraint belongs to a different component and removes exactly one coordinate; hence the constraints are independent. This proves (7.8), including the lower bound from the private coordinates. $\square$

The distinction between the coordinate change and the reconstruction map is central. The first is allowed to use the full presentation P because its task is to establish relative genericity. The second will use only finitely much *authorized* information. Giving the reconstruction map access to P would make the core upper bound far too weak.

7.3 Truncating only authorized secrets

For $C \neq \emptyset$, define the finite authorized join

$$B_{C,m} = \bigoplus_{\substack{\emptyset \neq S \subseteq C \\ k < m}} A_{S,k}. \quad (7.9)$$

Every such $B_{C,m}$ belongs to $\mathcal{I}_C$ by monotonicity and closure under finite joins. Its lower cone increases with m , and

$$\bigcup_m \text{Low}(B_{C,m}) = \mathcal{I}_C. \quad (7.10)$$

For the reverse inclusion in (7.10), it is enough that the terms with $S = C$ already generate $\mathcal{I}_C$.

Let $Z_S^{[m]}$ be the first- m -column truncation of Z_S . Define a map on the free generic tuple by

$$F_{C,m}(V) = F_C(V, (Z_S^{[m]})_{\emptyset \neq S \subseteq C}). \quad (7.11)$$

We identify its finite product domain with $2^{\mathbb{N}}$. The map is total and $B_{C,m}$ -computable. By Lemma 3.1 it preserves coarse similarity in the variable V .

Lemma 7.3 (The quantitative approximation bound). *The normal form satisfies*

$$\delta(X_C, F_{C,m}(V_C)) \leq \frac{2^{|C|} - 1}{|C|L} 2^{-m} \leq 2^{-m}. \quad (7.12)$$

Proof. For a fully observed component $S \subseteq C$, substituting $Z_S^{[m]}$ for Z_S changes only its last reconstructed stream. The other $|S| - 1$ streams are copied masks. The changed last stream differs on exactly $Z_S \Delta Z_S^{[m]}$,

which has upper density at most 2^{-m} . This description includes singleton components, whose sole stream is the last stream. Properly observed components and private streams do not change. There are $2^{|C|} - 1$ fully observed components among $|C|L$ output streams. Apply Lemma 3.1(c). The coefficient is at most one because each fully observed component contributes a distinct output stream. $\square$

There is no appeal here to cancellation among different error sets. Each changed stream has its own deterministic upper bound, and only finitely many streams are involved. This is why a single truncation level works for all authorized components of a fixed finite coalition.

8 Proof of simultaneous realization

Proposition 8.1 (Every intended generator survives). *For the construction above, $\mathcal{I}_C \subseteq \text{Core}(X_C)$ for every nonempty $C \subseteq [n]$.*

Proof. Fix a coarse description D of X_C . Extract from it the streams belonging to the all-of- C component. Each extracted stream is a coarse description of its exact share, by Lemma 3.1(a). The exclusive-or sum of these descriptions is a coarse description of Z_C , by Lemma 3.1(d) and (7.2). Theorem 5.2 therefore implies that D computes each member of $\mathcal{I}_C$. As this works for every D , the stated inclusion follows. $\square$

Proposition 8.2 (No unintended information survives). *For the same construction, $\text{Core}(X_C) \subseteq \mathcal{I}_C$ for every nonempty $C \subseteq [n]$.*

Proof. Use the fixed generic tuple V_C of Lemma 7.2, with backgrounds $B_{C,m}$ and maps $F_{C,m}$ from (7.11). Since $B_{C,m} \leq_T P$, the tuple V_C is 1-generic relative to $B_{C,m}$. The maps are total, $B_{C,m}$ -computable, and preserve coarse similarity. The approximation errors tend to zero by Lemma 7.3. All hypotheses of Lemma 4.5 are satisfied. Consequently,

$$\text{Core}(X_C) \subseteq \bigcup_m \text{Low}(B_{C,m}) = \mathcal{I}_C.$$

For clarity, the individual quantifier argument is this. Given $A \in \text{Core}(X_C)$, choose its positive radius η , then choose m with $2^{-m} < \eta$. For every $D \approx V_C$, the output $F_{C,m}(D)$ lies within that radius. Hence $A \leq_T B_{C,m} \oplus D$ for every such D . The relative generic-core theorem forces $A \leq_T B_{C,m} \in \mathcal{I}_C$. No index uniformly recovering A from those different outputs is needed. $\square$

Proof of the realization equivalence in Theorem 1.1. Necessity is Lemma 3.2. For sufficiency, choose the generator presentations, codes, masks, and private streams as in Section 7. Propositions 8.1 and 8.2 give every nonempty equality in (1.1). The empty equality holds by convention. The same single master generic works for all coalitions: there are only finitely many coordinate changes to consider, and each preserves genericity by a theorem rather than by an extra construction requirement. The remaining nonattainment and effectiveness clauses are proved in the next section. $\square$

8.1 A fully displayed three-participant example

For $n = 3$, the prescribed data are

$$\mathcal{I}_1, \mathcal{I}_2, \mathcal{I}_3, \mathcal{I}_{12}, \mathcal{I}_{13}, \mathcal{I}_{23}, \mathcal{I}_{123},$$

subject only to the inclusions forced by subscripts. Let Z_S code I_S . Use masks $R_{12}, R_{13}, R_{23}, U, V$ and private streams T_1, T_2, T_3 , whose eight-way join is 1-generic over P . Each participant has five streams:

Participant	Private	Singleton	First pair	Second pair	Triple
1	T_1	Z_1	R_{12}	R_{13}	U
2	T_2	Z_2	$Z_{12} \triangle R_{12}$	R_{23}	V
3	T_3	Z_3	$Z_{13} \triangle R_{13}$	$Z_{23} \triangle R_{23}$	$Z_{123} \triangle U \triangle V$

Coalition $\{1, 2\}$ fully observes the singleton components 1 and 2 and the pair component 12. Its observations of 13, 23, and 123 are proper, so the normal form treats them as generic coordinates. There are ten observed streams, three full-component parity relations, and seven free streams. Truncating the three authorized codes changes at most three streams, with density distance at most $(3/10)2^{-m}$. Thus its core is precisely I_{12} , even when I_{123} contains substantially more information.

The full coalition observes fifteen streams and seven independent parity relations, leaving eight free streams, exactly the original master masks and private streams. Its core is I_{123} . Nothing in this calculation requires the three pair ideals to be equal or comparable with one another.

For example, one may prescribe all singleton ideals as Comp, prescribe $I_{12} = \text{Low}(A)$ and $I_{13} = \text{Low}(B)$ for arbitrary A, B , prescribe $I_{23} = \text{Comp}$, and take I_{123} to be any countable ideal containing $A \oplus B$. Every one of these choices is realized simultaneously.

9 Representative spectra, nonattainment, and jump bounds

9.1 The spectrum criterion, with its short proof

For binary targets, write $X \leq_{\text{nc}} Y$ if every coarse description of Y computes a coarse description of X . Write $X \leq_{\text{uc}} Y$ if one Turing functional maps every coarse description of Y to a coarse description of X . Mutual reducibility defines the respective equivalences. Let $\text{Spec}_r(X)$ be the set of ordinary Turing degrees of representatives coarsely equivalent to X , where $r \in \{\text{nc}, \text{uc}\}$. These spectra should not be confused with the coarse degrees themselves.

Lemma 9.1 (Spectrum identity). *For every binary X ,*

$$\begin{aligned} \text{Spec}_{\text{nc}}(X) &= \text{Spec}_{\text{uc}}(X) = \{\deg_T(D) : D \approx X\} \\ &= \{\mathbf{b} : (\exists D \approx X) \deg_T(D) \leq \mathbf{b}\}. \end{aligned} \tag{9.1}$$

The common lower cone of any of these spectra, expressed as a collection of sets, is $\text{Core}(X)$.

Proof. If $D \approx X$, then D and X have exactly the same coarse descriptions, so the identity functional gives uniform equivalence. Uniform equivalence implies nonuniform equivalence. If $Y \equiv_{\text{nc}} X$, apply $X \leq_{\text{nc}} Y$ to Y as a description of itself; it computes a $D \approx X$. These facts give all inclusions except upward closure of the description spectrum.

For that closure, let B compute a coarse description D of X . Replace $D(2^k)$ by $B(k)$ for every k and leave all other positions unchanged. The resulting E still describes X because the powers of two have density zero. Also $E \leq_T B$ and $B \leq_T E$ by reading those powers, so $\deg_T(E) = \deg_T(B)$. This proves (9.1). Intersecting lower cones over the description spectrum is exactly the definition of the core. $\square$

The lemma is included to make the later inference self-contained; the same spectrum identity is already in the repository synthesis [5].

Proposition 9.2 (When a least representative exists). *The common spectrum in (9.1) has a least degree if and only if some member of $\text{Core}(X)$ computes a coarse description of X .*

Proof. If $A \in \text{Core}(X)$ computes $D \approx X$, then $A \leq_T D$ by the core property, so $D \equiv_T A$. Every member of the common spectrum computes some description of X and hence computes A . Thus $\deg_T(A)$ is attained and least. Conversely, let a least degree be represented by $D \approx X$ using Lemma 9.1. Every other coarse description computes D by leastness. Thus $D \in \text{Core}(X)$ and D itself is a description. $\square$

Remark 9.3 (Allowing natural-number-valued representatives). These conclusions do not change if the coarse classes contain arbitrary total functions $\mathbb{N} \rightarrow \mathbb{N}$. If the target is binary, convert a description E to a binary one by assigning 1 exactly where E is 1 and assigning 0 elsewhere. This introduces no error at any previously correct position. The self-application argument still gives a binary description computable from every representative. Its degree can be represented by a set coding the graph of the function. The other inclusions and sparse coding are already witnessed by binary descriptions. Hence the same spectrum identity and the same leastness criterion hold in the larger universe.

9.2 Nonattainment in every nonempty coalition

Theorem 9.4 (Simultaneous absence of a least representative). *For the finite-profile construction, every nonempty $C \subseteq [n]$ has no P -computable coarse description and no least degree in any of the three spectra in (9.1).*

Proof. Choose $i \in C$. The private stream T_i is a periodic coordinate of X_C and is 1-generic relative to P . If a P -computable oracle coarsely described X_C , extracting this coordinate would give a P -computable coarse description of T_i , contradicting Lemma 4.2.

Every set in $\text{Core}(X_C) = \mathcal{I}_C$ is computable in P : it is reducible to a finite join of generators from the presentation. Hence no core member can compute a coarse description of X_C . Proposition 9.2 excludes a least degree. The natural-number-valued extension in the previous remark shows that enlarging the set of representatives cannot restore one. $\square$

Corollary 9.5 (Exact behavior of the infimum). *In the constructed profile, fix $C \neq \emptyset$. If $\mathcal{I}_C = \text{Low}(B)$ is principal, the degree of B is the greatest lower bound of the representative spectrum but is not in that spectrum. If $\mathcal{I}_C$ is nonprincipal, the spectrum has no greatest lower bound in the Turing degrees.*

Proof. The common lower cone of the spectrum is exactly $\mathcal{I}_C$ by Lemma 9.1. A greatest lower bound is precisely a greatest Turing degree in that ideal. An ideal has such a degree exactly when it is principal. In the principal case that degree cannot be attained, because attainment would make it least, contrary to Theorem 9.4. $\square$

The statement excludes a *least* element, not necessarily a *minimal* element. A minimal element need not be below the other elements. Nothing in the proof identifies the full spectrum closely enough to exclude minimal elements, and the repository's target C5 is not settled by this argument.

9.3 A common effective bound

Theorem 9.6 (Relative lowness of the entire realization). *Choose the master generic $G \leq_T P'$ as in Lemma 4.4. Then the full join $H = X_{[n]}$ satisfies*

$$H \equiv_T P \oplus G,$$

and hence $P <_T H <_T P'$ and $H' \equiv_T P'$. For every nonempty C there is also the relative bound

$$P <_T P \oplus X_C <_T P', \quad (P \oplus X_C)' \equiv_T P'. \quad (9.2)$$

Proof. All shares are computable from $P \oplus G$, so $H \leq_T P \oplus G$. Conversely, the full join contains every private stream and every raw mask: for an all-of- S component the first $|S| - 1$ shares are exactly its masks. Thus H computes G . Taking the exclusive-or of the complete shares in each component recovers its exact code Z_S . Exact decoding of the columns of all these codes recovers the full array (7.4) uniformly. Hence $P \leq_T H$, proving $H \equiv_T P \oplus G$. Lemma 4.4 gives the stated full-join conclusions.

For a nonempty coalition, $P \oplus X_C \leq_T P \oplus G \equiv_T H$, so $(P \oplus X_C)' \leq_T H' \equiv_T P'$. The reverse reduction follows from $P \leq_T P \oplus X_C$. The private stream of any member of C shows $X_C \not\leq_T P$, proving the lower strict inequality. If the upper one failed, $P \oplus X_C \equiv_T P'$ and jumping would again give $P'' \equiv_T P'$. $\square$

Completion of Theorem 1.1. The realization equivalence was proved in Section 8. Theorem 9.4 supplies its simultaneous nonattainment clause, and Theorem 9.6 supplies (1.2). All clauses refer to the same construction. $\square$

Remark 9.7 (What the effective bound does and does not provide). The argument is uniform at the level of the given presentation: a fixed P' -effective generic construction and fixed finite formulas produce the shares. This is not a computable algorithm that takes an arbitrary abstract ideal as a finite input. Also, X_C need not compute P , so $X'_C \equiv_T P'$ is not claimed for every proper coalition. The precise uniform statement is (9.2), with the background joined in.

10 Access structures with genuinely generic unauthorized views

The general profile theorem permits different nontrivial ideals for many coalitions. An access structure is the useful special case where the only desired cores are Comp and one fixed secret ideal. There is then a stronger realization: unauthorized views themselves, before any corruption, can be 1-generic relative to the entire secret presentation.

Definition 10.1. An *access structure* on $[n]$ is an upward-closed family $\mathcal{A} \subseteq \mathcal{P}([n])$ with $\emptyset \notin \mathcal{A}$. Its members are called *authorized*. The empty family is allowed. Let $\min \mathcal{A}$ be its inclusion-minimal members.

Theorem 10.2 (Generic access-structure realization). *Let $\mathcal{I}$ be a countable Turing ideal presented by P , and let $\mathcal{A}$ be a finite access structure. There are binary participant oracles $X_1, \dots, X_n$ such that, for every coalition C ,*

$$\text{Core}(X_C) = \begin{cases} \mathcal{I}, & C \in \mathcal{A}, \\ \text{Comp}, & C \notin \mathcal{A}. \end{cases} \quad (10.1)$$

For each nonempty unauthorized C , the exact oracle X_C is 1-generic relative to P . In particular,

$$\text{Low}(X_C) \cap \text{Low}(P) = \text{Comp}. \quad (10.2)$$

Every nonempty coalition class has no least representative. All shares can be chosen computable in P' , with $(P \oplus X_C)' \equiv_T P'$ and $P <_T P \oplus X_C <_T P'$ for nonempty C . When $\mathcal{A}$ is nonempty, the full join itself satisfies $H \equiv_T P \oplus G$ and the sharper full-join bound (1.2).

Proof. Let Z be the single code of the generators of $\mathcal{I}$, so $Z \equiv_T P$. For every $S \in \min \mathcal{A}$, create an all-of- S component with the same secret stream Z , using disjoint mask coordinates for different components. Every participant receives at least one private generic stream.

To keep all joins equally interleaved, let

$$d_i = |\{S \in \min \mathcal{A} : i \in S\}|, \quad L_* = 1 + \max_i d_i.$$

Give participant i its d_i component streams and exactly $L_* - d_i$ additional, distinct private generic streams. These counts are positive. All mask and private streams form one master tuple G , chosen 1-generic relative to P . Participant i 's oracle is their L_* -way join.

If C is unauthorized, it contains no $S \in \min \mathcal{A}$. Its observations of every component are therefore proper. By the same affine changes as in Lemma 7.2, its entire flattened view is a nonempty coordinate projection of a P -computable homeomorphic image of G . There are no fully observed secrets to reconstruct. Thus X_C itself is 1-generic relative to P . Theorem 3.6 with computable background gives its ordinary core Comp , and Lemma 4.3 gives (10.2).

If C is authorized, it contains at least one minimal authorized set S . The corresponding parity sum of coarse share descriptions yields a coarse description of Z . Thus $\mathcal{I} \subseteq \text{Core}(X_C)$. For the reverse inclusion, form the same coalition normal form, with one copy of Z for each fully observed component. Replace all those copies by $Z^{[m]}$. Only finitely many output streams change, each on a set of upper density at most 2^{-m} . The reconstruction map is computable in $B_m = \bigoplus_{k < m} A_k$ and preserves coarse similarity in its generic coordinates. Lemma 4.5 gives $\text{Core}(X_C) \subseteq \bigcup_m \text{Low}(B_m) = \mathcal{I}$.

Every nonempty coalition contains a private generic stream and therefore has no P -computable description. All members of its core belong to $\text{Low}(P)$, so Proposition 9.2 excludes a least degree. Choose $G \leq_T P'$ for the bounds. We have $X_C \leq_T P \oplus G$, while a private stream forces $X_C \not\leq_T P$. The proof of Theorem 9.6 yields the relative jump assertions. If $\mathcal{A} \neq \emptyset$, the full coalition sees a complete component, so recovers $Z \equiv_T P$, as well as all masks and private streams; hence $H \equiv_T P \oplus G$. If $\mathcal{A}$ is empty, all views are simply joins of private generic streams, and no recovery of P from H is claimed. $\square$

10.1 Thresholds and a worked example

For $1 \leq t \leq n$, take $\mathcal{A} = \{C : |C| \geq t\}$. Its minimal members are exactly the t -element subsets. Thus any threshold pattern is realized, and each party receives $\binom{n-1}{t-1}$ component streams plus one private stream. This count measures the width of this particular construction, not an optimality claim.

For a two-of-three structure, one can write the whole scheme as

$$\begin{aligned} X_1 &= \text{Join}_3(T_1, R_{12}, R_{13}), \\ X_2 &= \text{Join}_3(T_2, Z \triangle R_{12}, R_{23}), \\ X_3 &= \text{Join}_3(T_3, Z \triangle R_{13}, Z \triangle R_{23}). \end{aligned}$$

The six-way join of the T and R streams is generic over $P \equiv_T Z$. Each singleton view is generic over P and computes no noncomputable P -computable set. Each pair robustly recovers the ideal $\mathcal{I}$ through its complete pair component, and its core contains nothing else. The triple has the same core $\mathcal{I}$, even

though it observes all three redundant copies. Every nonempty view lacks a least-degree coarse representative.

10.2 Why this is not a practical cryptographic claim

Equation (10.2) is a precise statement about exact Turing reductions, and (10.1) is a precise statement about information surviving arbitrary density-zero errors. They do not assert finite-key security, computational indistinguishability, Shannon secrecy of a sampled finite distribution, resistance to finite leakage models, or efficient share generation. The streams may be noncomputable, and the construction of a suitable generic uses a jump oracle. Its relevance is mathematical: it separates exact information, robust nonuniform information, and the finite combinatorics of cooperation.

11 A boundary: leastness cannot be prescribed arbitrarily

The main theorem realizes every monotone ideal profile while making all nonempty spectra unattained. Replacing that uniform negative requirement by arbitrary positive and negative leastness flags introduces genuine compatibility conditions.

Proposition 11.1 (Least representatives are closed under finite joining). *Suppose the coarse spectra of X and Y have least degrees $\mathbf{a}$ and $\mathbf{b}$. Then the coarse spectrum of $X \oplus Y$ has least degree $\mathbf{a} \oplus \mathbf{b}$, and*

$$\text{Core}(X \oplus Y) = \text{Low}(A \oplus B)$$

for representatives A, B of those two degrees.

Proof. By Proposition 9.2, choose coarse descriptions $D_X \approx X$ and $D_Y \approx Y$ with $D_X \equiv_T A$ and $D_Y \equiv_T B$, where $A \in \text{Core}(X)$ and $B \in \text{Core}(Y)$. Their join is a coarse description of $X \oplus Y$ computable from $A \oplus B$. Every description of $X \oplus Y$ projects to descriptions of X and Y , so it computes both A and B , hence $A \oplus B$. The joined description therefore has least degree $\mathbf{a} \oplus \mathbf{b}$. The core is at least $\text{Low}(A \oplus B)$ by this recovery and at most that ideal because this particular description has exactly that degree. $\square$

For a participant family, let $\mathcal{L}$ contain exactly the coalitions whose spectra have a least degree. It contains the empty coalition. If $C, D \in \mathcal{L}$, then $C \cup D \in \mathcal{L}$: joining their views and then deleting duplicated coordinates is uniformly coarse-equivalent to the view of the union. Moreover, the least degree at the union must be the join of the two least degrees. Thus $\mathcal{L}$ must be union-closed, and its principal core labels must satisfy this join identity. These conditions are not imposed on coalitions outside $\mathcal{L}$.

This explains why an example of arbitrary synergy cannot be obtained by using individually coarsely computable shares. A finite join of computable coarse descriptions is again computable. In the two-share construction the individual cores are computable, but no computable member of either core computes a description of its share. The gap between having a principal core and attaining its greatest degree is exactly what makes arbitrary joint information possible.

12 Proof audit and reproducible finite verification

12.1 Logical dependency chain

The main proof can be checked in the following order. The only external coarse-computability inputs are Theorems 3.3 and 3.6. The first yields the robust-radius corollary. Finite counting gives the block and column calculations, establishing the single-ideal identity. The two inputs together yield the transfer lemma. Finite affine algebra and elementary genericity invariance yield the coalition normal form. That form and the explicit approximation bound feed the transfer lemma and give the upper inclusion. Decoding one fully observed component gives the lower inclusion independently. Private generic streams then give nonattainment, and exact access to the full array gives the jump bound.

In particular, the proof of the upper inclusion does not use the main realization theorem, the desired ideal equality, the absence of a least representative, or any claim that the output is minimal. There is no self-reference through the intended conclusion.

Potential failure	Check or repair used here
Coarse information confused with exact information	$Z_S \equiv_T P_S$ is an exact statement; description decoding is only per-generator and nonuniform.
Ordinary genericity used with an unauthorized oracle	The master tuple is generic relative to the entire presentation P ; the reconstruction step uses only $B_{C,m}$.
Separately generic masks assumed jointly generic	All masks and padding streams are projections of one master generic.
Unauthorized secrets left in the final reduction oracle	They enter an affine coordinate change, not the truncated reconstruction map.
Genericity assumed under arbitrary Turing equivalence	Only explicit homeomorphisms and finite open coordinate projections are used.
A coarse description extracted from a zero-density channel	All extraction channels here have fixed positive periodic or dyadic density.
An infinite union of zero-density error sets used	Every decoding or perturbation comparison uses only finitely many channels at once. Infinite columns are controlled by the explicit tail density 2^{-m} .
Finite majority correction treated as uniformly available	The correction is hard-coded separately for each description and generator; no uniform exact decoder is asserted.
No least element confused with no minimal element	The proof uses the leastness criterion only. Minimal elements remain unclassified.
Relative lowness confused with absolute lowness	The exact bounds are $H' \equiv_T P'$ and $(P \oplus X_C)' \equiv_T P'$.
Finite testing advertised as theorem verification	The Python tests check only the finite algebra and counting listed below.

12.2 The finite checks actually run

The archive contains `checks/verify_finite.py`, requiring only the Python standard library. It writes `checks/results.json`. The run included with this article passed all checks in the following table.

[write] In this report the program is shipped as `code/verify_finite.py` and the recorded run as `data/results.json`. The program writes `results.json` beside itself, so run it on a copy (see the report's `README.md`); run in place it would add `code/results.json`. At intake a run on a copy passed and reproduced the recorded file apart from line endings.

Check	Executed scope
All-of- r exclusive-or reconstruction	$1 \leq r \leq 8$; 510 cases
Proper observations have identical finite distributions for either secret bit	502 observed-subset comparisons
Coalition mask ranks and independent secret constraints	All 247 nonempty coalitions for $1 \leq n \leq 7$
Affine normal-form reconstruction and inverse	69,888 seeded cases, $n \leq 6$
Exact dyadic tail formula	22,539 pairs (N, m)
Column-index inversion	50,000 positions
Exclusive-or error inclusion	5,460 exhaustive bit-tuple pairs
Majority correctness and necessary error count	131,628 cases; block length at most 16
Recovery of finite monotone access families from minimal members	All 167 four-participant families with unauthorized empty coalition

The rank test represents the masks as independent formal coordinates over $\mathbb{F}_2$. For each coalition it checks

$$\text{rank}(\text{mask observation matrix}) = |C|L - (2^{|C|} - 1).$$

It also checks that adjoining the formal secret coordinates makes the observation rows independent. Together these tests support the finite normal-form bookkeeping. They do not test genericity of an infinite sequence, recovery from every density-zero perturbation, or any oracle jump equality. Those are proved in the article, using the declared inputs.

The fixed seed makes the non-exhaustive reconstruction checks reproducible; the distribution, rank, tail, majority, and access-family checks are exhaustive over their stated finite ranges. No statistical confidence claim is attached to the seeded checks.

12.3 A route to a genuine formalization

A faithful implementation should first define binary coarse descriptions, periodic joins, and dyadic column extraction over the repository's actual oracle semantics. The elementary density lemmas and the block decoder can then be formalized without any genericity machinery. The decoder's output should be an existential Turing reduction, with its finite correction visible; a uniform decoder theorem would be false.

The ideal code should be represented by a two-parameter array, keeping the array oracle separate from the ideal generated by its finite joins. The finite affine part can be proved independently over vectors over $\mathbb{F}_2$, then lifted pointwise to streams. Useful intermediate theorem interfaces are: invertibility of the missing-mask replacement, preservation of genericity by an effective homeomorphism and a finite projection, and coarse-similarity preservation by reconstruction.

The two published inputs can initially be explicit theorem parameters in a conditional development. Such a development would establish a correct implication from those inputs but would not be an assumption-free proof of the final theorem. A complete formalization must either prove the inputs or import already proved declarations whose dependencies have been audited. The last layers are the transfer lemma, finite coalition assembly, nonattainment criterion, and the ordinary generic jump calculation. The finite code shipped here can generate examples for this effort, but it is not itself a proof-producing formalizer.

13 Further research questions and topics

The questions below are proposed continuations of this article. Unless a repository label is explicitly mentioned, they are not claimed to be published open problems. They separate issues actually settled above from those that require additional ideas.

13.1 Full proof verification and priority

Question 13.1 (Formalize the finite-profile theorem). Can the complete theorem, including its two published inputs, be added to ProveIt’s genuine oracle-semantics development with an explicit axiom audit and no admitted mathematical statements?

A useful first milestone is the complete finite affine and density layer; a second is the single-ideal identity; the critical third is the relative generic-core and compactness interface. A literature comparison should run in parallel, searching for equivalent ideal-valued or mass-problem formulations rather than only the terminology “coarse core.” A successful priority search may reclassify the result as a synthesis; it would not invalidate the construction or its proofs.

13.2 Countably many participants

Question 13.2 (A coherent countable realization theorem). Given countable Turing ideals assigned monotonically to all finite subsets of $\mathbb{N}$, with computable empty ideal, can they be realized by one countable family $(X_i)_{i \in \mathbb{N}}$? Which additional conditions are needed if ideals are also assigned to infinite coalitions?

Applying the finite theorem independently to initial segments does not answer this: the resulting shares need not agree on common participants. A countable construction has infinitely many incident components and no single finite mask rank. One possible route is to put components into shrinking positive-density columns and prove a tail approximation lemma that eliminates all but finitely many observed components. The main danger is unauthorized information in the infinite collection of coordinate changes. Infinite coalition joins introduce a further distinction between uniform access to all participants and individual nonuniform reductions.

13.3 Extending fixed shares rather than rebuilding them

Question 13.3 (An extension property). Given already fixed $X_1, \dots, X_n$, which monotone extensions of their core profile to $n + 1$ participants are realizable while preserving those oracles exactly, or at least preserving their coarse-equivalence classes?

The present theorem rebuilds all shares from one master generic. It does not assert an amalgamation or extension property for arbitrary existing shares. Exact preservation may impose spectrum constraints invisible in the cores. Even the two-share question, with X fixed and only Y free, asks for more than Theorem 6.1. It would be useful to identify a class of extendible generic presentations and a precise obstruction outside that class.

13.4 Prescribing attainment together with the ideals

Question 13.4 (Mixed leastness profiles). Characterize the pairs consisting of a finite ideal profile $(\mathcal{I}_C)$ and a family $\mathcal{L}$ of coalitions whose spectra must have least degrees. Are principality on $\mathcal{L}$, union

closure of $\mathcal{L}$, and the join identities of Proposition 11.1 sufficient, or are there further compatibility conditions?

Our construction settles the case $\mathcal{L} = \{\emptyset\}$. Positive attainment cannot simply be enforced by adding a least-degree block code: that may expose information to larger coalitions and can conflict with prescribed nonattainment there. A promising finite starting case is three participants with only one singleton and one overlapping pair required to attain their cores.

13.5 The full spectra and the repository's C5

Question 13.5 (Beyond common information). What are the complete coarse representative spectra of the two-share code and the finite-profile construction? Can the generic masks be chosen so that selected spectra have no minimal element, while preserving every core equality and the common jump bound?

This directly touches C5 of the repository plan [4], but the present no-least theorem does not answer it. The core gives the common lower cone, not the collection of degrees which compute a description. Progress would require an effective description-building or description-lowering argument, not just a further common-information calculation. A first target is the two-share case for one principal ideal, where the intended infimum is completely explicit.

13.6 Degree bounds independent of a chosen enumeration

Question 13.6 (Presentation-sensitive complexity). For a fixed abstract countable ideal, what is the least possible complexity of a presentation admitting the bounds proved here? Can one characterize when a given oracle Q bounds a realization with the required exact profile and nonattainment?

The theorem gives a uniform upper bound from P' , not an optimal bound for the ideal itself. Rearranging a generating sequence can add information to its presentation without changing the ideal. A useful invariant would separate the cost of presenting the ideal from the cost of producing a generic extension. Asking to construct these particular nonattained realizations below the same P is impossible because they intentionally have no P -computable descriptions; meaningful improvements must alter the presentation bound or the construction requirements.

13.7 Uniform reductions and other description notions

Question 13.7 (Uniform coarse structure of the profile). Can one prescribe, along with the core profile, selected uniform coarse reducibilities among coalition views? Which finite patterns are compatible with the nonuniform recoveries used here?

Proposition 5.3 settles only literal uniform *exact* recovery. It says nothing comparable about uniform production of descriptions. The component parity maps are uniform at the description level, whereas block correction is not uniform at the exact-set level. Keeping these two interfaces separate may lead to a useful classification of uniform coarse access structures. Effective-dense descriptions, which permit explicit omissions but no wrong answers, require a separate study; the density-zero error argument cannot be transferred merely by changing terminology. This is relevant to C2 and C8 of [4] without claiming to resolve them.

13.8 Efficient finite architectures

Question 13.8 (Stream complexity and linear sharing). For a prescribed finite ideal profile, what is the smallest number of positive-density streams per participant in a realization of this form? Can the all-subset construction be replaced by a smaller linear sharing architecture while preserving the generic normal form?

The construction uses $2^{n-1} + 1$ streams per participant. That is an explicit bound, not a lower bound. For access structures, minimal authorized sets already reduce the number of components. More efficient finite linear schemes are candidates, but an adaptation must prove that an unauthorized observation is a projection of an effective homeomorphic change of masks, and must give a density-continuity estimate for reconstruction. Finite rank alone should not be substituted for these oracle-level assertions.

13.9 Other error ideals and higher reducibilities

Question 13.9 (Replacing asymptotic density zero). For which ideals of allowed error sets does a finite monotone-profile realization theorem hold? What changes for upper Banach density zero, logarithmic density zero, or a fixed summable ideal?

The proof separates three required ingredients: a robust code on channels from which descriptions can be extracted; a finite-operation error calculus and controlled tails; and an analogue of the robust-radius plus generic-core mechanism. The finite sharing algebra survives unchanged, but the compactness and coding properties do not follow from abstract ideal closure. A carefully chosen error ideal may therefore supply either a generalization or a structural counterexample.

Question 13.10 (Hyperarithmetic profiles). Can arbitrary finite monotone profiles of suitable countable hyperdegree ideals be realized as information common to all coarse descriptions, with hyperarithmetic rather than Turing reducibility?

The repository's report 10 discusses a failure of the corresponding hyperarithmetic robust-radius principle [7]. Consequently the transfer lemma must not simply be reprinted with $\leq_h$ in place of $\leq_T$. One needs a replacement for the step which turns recovery from all coarse descriptions into recovery from nearby truncated reconstructions. The finite affine normal form may still be useful, but it does not solve that missing higher-computability problem.

14 Conclusion

The core records common robust information, not the difficulty of choosing a representative and not all information present in exact oracles. The finite realization theorem shows that its interaction with cooperation is as unrestricted as monotonicity permits: every finite monotone diagram of countable Turing ideals can be realized, even while every nonempty representative spectrum fails to attain a least degree.

The key proof is the coalition upper bound. A generic normal form isolates all fully observed codes; their density truncations require only finitely many authorized generators. Cone-avoiding compactness and the relative generic-core theorem then remove the free coordinates. This proves an exact ideal equality rather than merely showing that some chosen secret is recoverable. The same

construction has a common relative low jump, and the access-structure variant makes all unauthorized exact views generic over the entire presentation.

The single-ideal theorem verifies the repository’s C4 sketch rather than superseding its provenance. The proposed new contribution is the simultaneous finite classification and its refinements. Full spectra, minimal elements, coherent countable extensions, and mixed attainment patterns remain separate research targets. The proofs and finite checks here provide explicit objects and interfaces with which to attack them.

A Notation and construction index

Notation	Meaning
$X \triangle Y$	Symmetric difference / pointwise exclusive or.
$X \oplus Y$	Oracle join; throughout, a specified equal-density interleaving.
$\delta(X, Y)$	Upper asymptotic density of the disagreement set.
$D \approx X$	$\delta(D, X) = 0$; D is a coarse description of X .
$\text{Core}^B(X)$	Sets computed by $B \oplus D$ for every $D \approx X$.
$\text{Core}(X)$	The unrelativized core, with computable background.
$\text{Low}(B)$	The Turing ideal $\{A : A \leq_T B\}$.
$J(A)$	Redundant block code, constant on $[2^j, 2^{j+1})$.
R_k	Positive-density column $\{2^k(2t + 1) - 1 : t \in \mathbb{N}\}$.
Z_S	Ideal code of the generator sequence $(A_{S,k})_k$.
$Z_S^{[m]}$	First- m -column truncation; the remaining columns are zero.
P	Oracle presenting all generator sequences; not assumed to belong to any prescribed ideal.
G	Single master tuple of masks and private streams, 1-generic relative to P .
$W_{S,i}$	Participant i ’s share in the all-of- S component.
T_i	A private generic stream ensuring nonattainment.
X_C	View of coalition C , the finite join of its participant oracles.
V_C	Free generic coordinates of the coalition normal form.
$B_{C,m}$	Finite join of generators in authorized components $S \subseteq C$ with $k < m$.
$F_{C,m}$	Reconstruction with authorized secrets truncated to m columns.
H	Full join $X_{[n]}$, equivalent to $P \oplus G$ in the main construction.

B Source review and reproducibility details

The repository comparison used the pinned commit displayed in Section 2, not an unversioned assumption about future contents. The inspected core documents were the TuringDegrees README, the coarse research plan’s C1–C8 discussion, the synthesis’s core and spectrum sections and result inventory, and report 10’s README. The literature comparison used the HTML text of the two arXiv papers in the bibliography, the author-hosted publication record for [1], and the publisher record for [3]. The latter was used for attribution of the classical access-structure architecture, not as an uninspected source for any computability theorem.

The proof uses the theorem numbering of the cited arXiv versions. All newly derived claims are proved in this article. No claim is made that the repository was exhaustively audited file by file or that

the literature search was exhaustive. Search results with no relevant mathematical content were not treated as evidence of originality.

The source is a standalone $\text{\LaTeX}$ file with an embedded bibliography. The build script runs $\text{pdf\LaTeX}$ three times to resolve references and contents. It requires the New PX font packages and standard AMS, geometry, microtype, booktabs, enumitem, xurl, titlesec, and hyperref packages available in common TeX Live installations. No font binaries are included in the archive. Run the finite checks separately with `python3 checks/verify_finite.py`. The supplied PDF was compiled from this source; document-layout checks are recorded in the accompanying proof audit rather than confused with mathematical verification.

[write] The supplied 28-page PDF is not shipped with report 11; `article.pdf` is a build of this text, which adds the [write] material listed in Section 2.5. The delivered `build.sh` is shipped as `code/build.sh`; it changes to its own directory and compiles `article.tex` there, so it does not build the report in place. Build with `latexmk` as the report’s `README.md` describes. The finite checks are `code/verify_finite.py`, and the layout record is `data/validation.json`; it describes the delivered PDF.

References

- [1] D. R. Hirschfeldt, C. G. Jockusch, Jr., R. Kuyper, and P. E. Schupp, *Coarse reducibility and algorithmic randomness*, *Journal of Symbolic Logic* **81** (2016), 1028–1046. Theorem 3.7 (cone-avoiding compactness) and Theorem 4.2 (generic core). Preprint: <https://arxiv.org/abs/1505.01707>. Text checked: <https://arxiv.org/html/1505.01707v1>. Publication record: <https://www.math.uchicago.edu/~drh/Papers/coarsereducibility.html>.
- [2] D. R. Hirschfeldt, C. G. Jockusch, Jr., and P. E. Schupp, *Coarse computability, the density metric, Hausdorff distances between Turing degrees, perfect trees, and reverse mathematics*, arXiv:2106.13118v1 (2021). Lemma 3.10 and the proof of Theorem 3.11(2). <https://arxiv.org/html/2106.13118v1>.
- [3] M. Ito, A. Saito, and T. Nishizeki, *Secret sharing scheme realizing general access structure*, *Electronics and Communications in Japan (Part III: Fundamental Electronic Science)* **72**, no. 9 (1989), 56–64. <https://doi.org/10.1002/ecjc.4430720906>.
- [4] V. Reshetnikov, ProveIt repository, *Turing Degrees: A Unified Research Report*, second edition, 18 September 2026, Section “Coarse and effective-dense representative spectra,” especially target C4 and Proposition “proposed.” Repository snapshot 1a1396d4d3a2ac6812692df520517d86ba3a4785. https://github.com/VladimirReshetnikov/ProveIt/blob/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/Research/CoarseDegrees/research-plan/turing_degrees_unified.tex.
- [5] V. Reshetnikov, ProveIt repository, *Coarse Classes Without Least Turing Degrees: Deduplicated results of the nine research reports on target C1*, 18 September 2026. Same repository snapshot. Conventional research synthesis; status and provenance distinctions are retained in the present article. https://github.com/VladimirReshetnikov/ProveIt/blob/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/Research/CoarseDegrees/research-synthesis/Turing_Degrees_Synthesis.tex.

- [6] V. Reshetnikov, ProveIt repository, *Turing degrees in Lean and Rocq*, project README, same snapshot. <https://github.com/VladimirReshetnikov/ProveIt/blob/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/README.md>.
- [7] V. Reshetnikov, ProveIt repository, *Coarse Classes, Hyperdegrees, and the Cone Filter*, research report 10, README dated 18 September 2026 with additions of 19 September 2026, same snapshot. Cited only for its stated higher-computability research boundary; not used as an input to any theorem here. <https://github.com/VladimirReshetnikov/ProveIt/blob/1a1396d4d3a2ac6812692df520517d86ba3a4785/Computability/TuringDegrees/Research/CoarseDegrees/research-reports/10/README.md>.